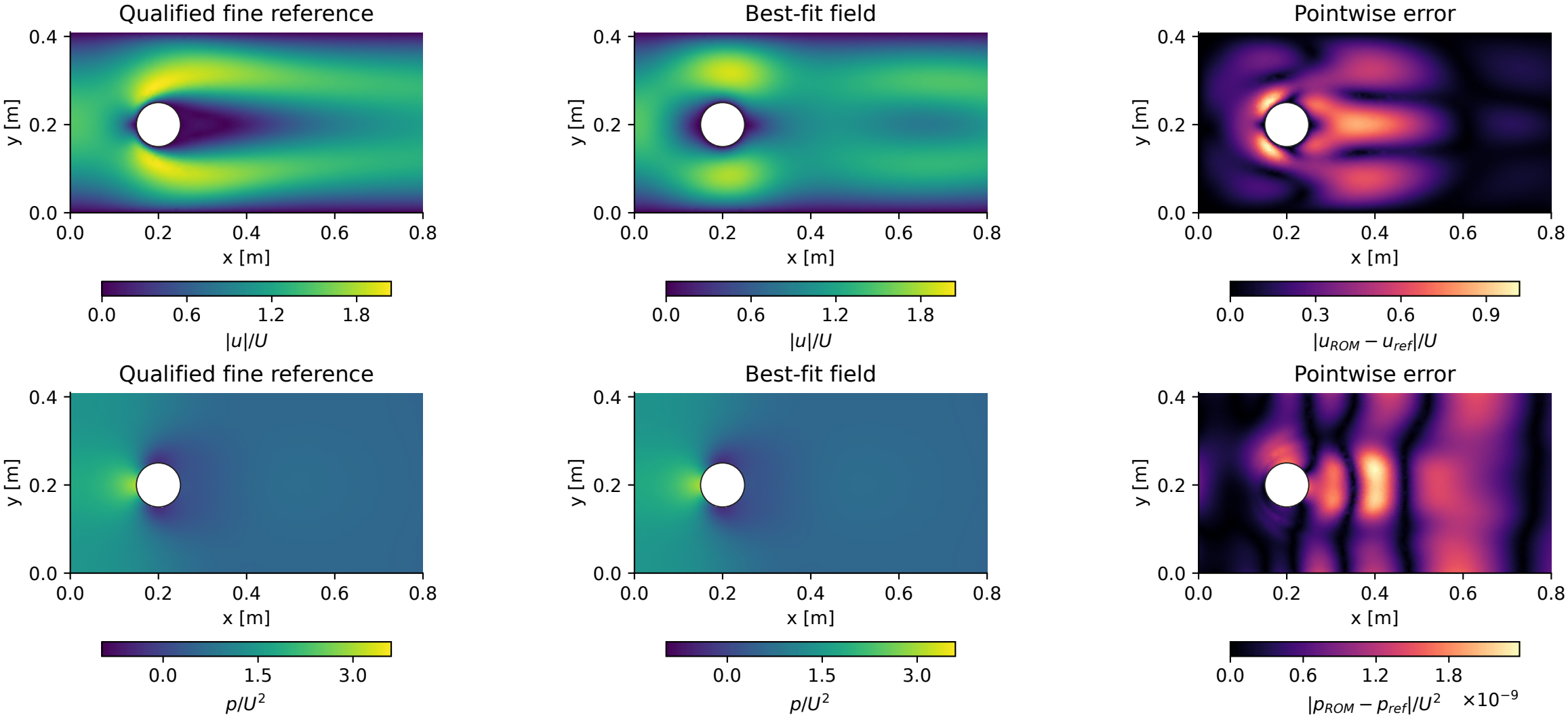

Neural bank: best-fit reconstruction — Re = 23.125
Velocity (top) and pressure (bottom); near-cylinder view
Qualified fine reference
Best-fit field
Pointwise error
y [m]
x [m]
0.0
0.2
0.4
0.6
0.8
0.0
0.6
1.2
1.8
|u|/U
0.0
0.3
0.6
0.9
|u_ROM − u_ref|/U
Qualified fine reference
Best-fit field
Pointwise error
0.0
1.5
3.0
p/U²
0.0
0.6
1.2
1.8
|p_ROM − p_ref|/U²
×10⁻⁹
Eight neural velocity modes plus shared pressure/stability spaces. Not an online flow solve.
Display interpolation of saved samples; error scales differ between figures. Development cases, not validation.